About the Dataset

- **Type:** Retail / E-commerce transactional dataset
- **Nature:** Structured (tabular data)
- **Level:** Customer transaction level

Key Features:

- **Numerical:** Age, Purchase Amount, Previous Purchases
- **Categorical:** Product Category, Payment Method, Season, Gender, Subscription Status, Shipping Type, Discount Applied

👉 This dataset captures detailed customer interactions and purchasing behavior, enabling in-depth business analysis and decision-making.

Domain Background

In the highly competitive **e-commerce industry**, understanding customer behavior is essential for driving revenue and improving retention.

Businesses continuously track:

- What customers buy
- How often they purchase
- How much they spend
- How they respond to discounts and marketing campaigns

Despite having large volumes of data, many companies struggle to convert raw data into **actionable insights**, leading to:

- Inefficient marketing spend
- Poor customer retention
- Missed revenue opportunities

Why This Analysis Matters

This analysis helps businesses:

- Identify **high-value and loyal customers**
- Understand **purchase patterns across categories and segments**
- Evaluate the effectiveness of **discount strategies**
- Improve **customer retention and engagement**
- Increase **Customer Lifetime Value (CLV)**
- Make **data-driven decisions to optimize revenue**

Project Objective

The goal of this project is to transform raw customer transaction data into **actionable business insights** that can help:

- Optimize marketing strategies
- Improve customer retention
- Enhance revenue generation
- Support strategic decision-making

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🔷 Title: Data Understanding & Analytical Approach

🔷 Section 1: Dataset Overview

The dataset contains transactional and customer-level information used to analyze purchasing behavior, discount impact, and revenue patterns.

Key Features:

- Customer ID
- Age
- Gender
- Purchase Amount
- Discount Applied
- Product Category
- Shipping Type
- Purchase Date

🔷 Section 2: Data Cleaning & Preparation

Before analysis, the dataset was cleaned and transformed to ensure accuracy and consistency.

Steps performed:

- Removed duplicate records
- Handled missing/null values
- Standardized categorical values (e.g., Yes/No, Male/Female)
- Converted data types (date, numeric fields)
- Created derived columns (age groups, revenue segments)

🔷 Section 3: Analytical Approach

The analysis was structured to answer key business questions:

- Who are the **high-value customers**?
- How do **discounts impact purchasing behavior**?
- Which **customer segments drive maximum revenue**?
- What is the **effect of shipping type on order value**?
- Are there identifiable **trends across age groups and categories**?

🔷 Section 4: Tools & Technologies Used

The following tools were used throughout the project:

- **Python (Pandas)** → Data cleaning
- **SQL** → Data extraction & analysis
- **Power BI** → Dashboard & visualization

Data Cleaning & Preparation (Python)

Data Cleaning & Preprocessing (Python - Pandas)

Data cleaning and pre processing were performed using Python (Pandas) to ensure data consistency, handle missing values, and prepare the dataset for accurate analysis.

◈ 1. Initial Data Exploration

The dataset was explored to understand its structure and quality.

- Checked dataset shape and size
- Reviewed column data types
- Generated summary statistics
- Identified missing values

◈ 2. Data Standardization

Product categories were inconsistent and were corrected using mapping logic.

- Standardized **Category** column based on *Item Purchased*
- Ensured uniform categorization across:
  - Clothing
  - Electronics
  - Accessories
  - Footwear

◈ 3. Handling Missing Values

Missing values were treated using logical and category-based imputation:

- **Size column:**
  - Electronics → “Not Applicable”

    - Accessories → “Free Size”
    - Footwear → “Not Available”
    - Clothing → filled using **mode**
- **Review Rating:**
    - Filled using **average rating per product**
- **Previous Purchases:**
    - Missing values replaced with **0**
- **Purchase Amount:**
    - Filled using **mean purchase amount per item**

◈ 4. Duplicate Removal

Duplicate records were identified and removed to ensure data accuracy.

- Checked duplicate entries
- Removed duplicates based on **Customer ID**

◈ 5. Feature Engineering & Formatting

Data was transformed to improve usability:

- Renamed columns (lowercase, removed spaces)
- Rounded review ratings for consistency

Small code snippet

```
df['Category'] = df['Item Purchased'].map(correct_mapping)
df['Review Rating'] = df['Review Rating'].fillna(
    df.groupby('Item Purchased')['Review
Rating'].transform('mean')
)
df.drop_duplicates(subset=['Customer ID'], inplace=True)
```

◈ Data Quality Outcome

After pre-processing:

- Dataset became **free from duplicates and major missing values**
- Categories were **standardized for accurate grouping**
- Numerical fields were **consistent and analysis-ready**.

Data Analysis Using SQL

Insight 1: Revenue Concentration (Pareto Insight)

◈ Objective:

To analyze how revenue is distributed across customers and identify whether a small group contributes to the majority of revenue.

◈ SQL Query:

```
WITH customer_revenue AS (
    SELECT
        customer_id,
        SUM(purchase_amount) AS total_spent
    FROM customer_behaviour_analysis
    GROUP BY customer_id
),

ranked AS (
    SELECT
        customer_id,
        total_spent,
        SUM(total_spent) OVER (ORDER BY total_spent DESC) AS
cumulative_revenue,
        SUM(total_spent) OVER () AS total_revenue
    FROM customer_revenue
),

flagged AS (
    SELECT *,
        (cumulative_revenue / total_revenue) AS cum_pct
    FROM ranked
)
```

```
SELECT
    COUNT(*) AS customers_contributing_80_percent,
    (SELECT COUNT(DISTINCT customer_id) FROM
customer_behaviour_analysis) AS total_customers,
    ROUND(
        COUNT(*) * 100.0 /
        (SELECT COUNT(DISTINCT customer_id) FROM
customer_behaviour_analysis),
    2) AS percentage_customers
FROM flagged
WHERE cum_pct <= 0.8;
```

◈ Result:

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

customers_contributing_80_percent	total_customers	percentage_customers
1430	5000	28.60

✍ Final Insight

Approximately **28.6% of customers contribute to 80% of total revenue**, indicating a strong revenue concentration among a smaller segment of customers.

🔥 Business Impact

This suggests that focusing on high-value customer retention strategies can significantly improve revenue efficiency and long-term profitability.

Insight 2: Discount vs Purchase Behavior

◈ Objective

To analyze how discount application impacts customer purchasing behavior in terms of order volume and average purchase value.

◈ SQL Query:-

```
SELECT
    discount_applied,
    COUNT(*) AS total_orders,
    COUNT(DISTINCT customer_id) AS unique_customers,
    ROUND(AVG(purchase_amount), 2) AS avg_purchase_value,
    ROUND(SUM(purchase_amount), 2) AS total_revenue
FROM customer_behaviour_analysis
GROUP BY discount_applied;;
```

◆ Result:

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

discount_applied	total_orders	unique_customers	avg_purchase_value	total_revenue
No	2756	2756	182.24	502241.64
Yes	2244	2244	219.22	491940.78

Non-discounted transactions generated slightly higher total revenue due to a greater number of orders.
However, discounted transactions showed a **higher average purchase value (219.22 vs 182.24)**.

Final Insight:

Customers who receive discounts tend to spend more per transaction, even though the overall number of orders is lower compared to non-discounted purchases.

Business Impact:

Discounts appear to encourage higher spending per order, suggesting they can be used strategically to increase average order value. However, since total revenue remains similar, discounts should be targeted rather than applied broadly.

Insight 3: Age Group Contribution

◆ Objective

To analyze how different age groups contribute to overall revenue and identify high-value customer segments.

◈ SQL Query:-

```
SELECT
  CASE
    WHEN age BETWEEN 18 AND 25 THEN '18-25'
    WHEN age BETWEEN 26 AND 35 THEN '26-35'
    WHEN age BETWEEN 36 AND 50 THEN '36-50'
    ELSE '51+'
  END AS age_group,
  COUNT(*) AS total_orders,
  COUNT(DISTINCT customer_id) AS unique_customers,
  ROUND(AVG(purchase_amount), 2) AS avg_purchase_value,
  ROUND(SUM(purchase_amount), 2) AS total_revenue,
  ROUND(SUM(purchase_amount) * 100.0
  / SUM(SUM(purchase_amount)) OVER (), 2) AS revenue_percentage
FROM customer_behaviour_analysis
GROUP BY age_group
ORDER BY total_revenue DESC;
```

Result:

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

age_group	total_orders	unique_customers	avg_purchase_value	total_revenue	revenue_percentage
51+	1917	1917	208.32	399345.32	40.17
36-50	1408	1408	189.42	266706.7	26.83
26-35	935	935	189.76	177421.93	17.85
18-25	740	740	203.66	150708.48	15.16

The **51+ age group contributes the highest share of revenue (40.17%)**, followed by the 36–50 group (26.83%).
Younger segments (26–35 and 18–25) contribute comparatively less to total revenue.

Insight:

Older customers (51+) are the most valuable segment in terms of revenue contribution, indicating higher purchasing power and consistent spending behavior.

Business Impact:

The company should prioritize this segment with premium offerings, loyalty programs, and personalized experiences, while using targeted promotions to engage younger age groups.

Insight 4: Shipping Type Impact

◈ Objective

To analyze how different shipping types influence customer purchasing behavior and revenue generation.

◈ SQL Query:-

```
SELECT
    shipping_type,
    COUNT(*) AS total_orders,
    COUNT(DISTINCT customer_id) AS unique_customers,
    ROUND(AVG(purchase_amount), 2) AS avg_purchase_value,
    ROUND(SUM(purchase_amount), 2) AS total_revenue
FROM customer_behaviour_analysis
GROUP BY shipping_type
ORDER BY total_revenue DESC;
```

Result:

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

shipping_type	total_orders	unique_customers	avg_purchase_value	total_revenue
Express	1199	1199	358.12	429380.98
Standard	1201	1201	340.57	409020.45
Free Shipping	675	675	60.41	40777
Store Pickup	650	650	59.89	38931
2-Day Shipping	627	627	60.73	38080
Next Day Air	648	648	58.63	37993

Express shipping generates the highest total revenue (429,380.98) and also has the **highest average purchase value (358.12)**.
Standard shipping follows closely in both order volume and revenue, while other shipping types contribute significantly less.

Insight:

Customers who choose faster delivery options (like Express and Standard) tend to spend significantly more per transaction compared to other shipping methods.

Business Impact:

Encouraging customers to opt for premium shipping (e.g., Express) through incentives or thresholds (like free express delivery above a certain amount) can significantly increase average order value and overall revenue.

Insight 5: Category Performance

◈ Objective

To analyze which product categories contribute the most to revenue and identify high-performing and low-performing segments.

◈ SQL Query:-

```
SELECT
    category,
    COUNT(*) AS total_orders,
    COUNT(DISTINCT customer_id) AS unique_customers,
    ROUND(AVG(purchase_amount), 2) AS avg_purchase_value,
    ROUND(SUM(purchase_amount), 2) AS total_revenue,
    ROUND(
        SUM(purchase_amount) * 100.0  / SUM(SUM(purchase_amount))
                                        OVER (), 2) AS revenue_percentage
FROM customer_behaviour_analysis
GROUP BY category
ORDER BY total_revenue DESC
LIMIT 5;
```

Result:

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

category	total_orders	unique_customers	avg_purchase_value	total_revenue	revenue_percentage
Electronics	637	637	763.53	486366.96	48.92
Accessories	1404	1404	138.79	194866.87	19.6
Clothing	1883	1883	97.29	183188.39	18.43
Footwear	752	752	147.92	111236.2	11.19
Outerwear	324	324	57.17	18524	1.86

Electronics is the top-performing category, contributing **48.92% of total revenue** with the **highest average purchase value (763.53)**.
It is followed by **Accessories (19.6%)** and **Clothing (18.43%)**, while **Outerwear contributes the least (1.86%)**.

Insight:

Revenue is highly concentrated in the Electronics category, which drives nearly half of total revenue, indicating a strong preference for high-value products. In contrast, categories like Outerwear generate minimal revenue despite contributing to order volume.

Business Impact:

The company should prioritize Electronics through targeted marketing, premium positioning, and inventory investment. Low-performing categories like Outerwear should be reevaluated through pricing strategies, bundling, or promotional efforts.

Insight 6: Customer Purchase Frequency

◈ Objective

To analyze how frequently customers make purchases and identify the proportion of repeat vs one-time buyers.

◈ SQL Query:-

```
WITH customer_orders AS (
    SELECT
        customer_id,
        COUNT(*) AS order_count
    FROM customer_behaviour_analysis
    GROUP BY customer_id
)
SELECT
CASE
        WHEN order_count = 1 THEN 'One-time Customers'
          ELSE 'Repeat Customers'
END AS customer_type,
COUNT(*) AS total_customers,
ROUND(COUNT(*) * 100.0 / SUM(COUNT(*)) OVER (), 2) AS percentage
FROM customer_orders
GROUP BY customer_type;
```

Result:

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

customer_type	total_customers	percentage
One-time Customers	5000	100.00

All customers in the dataset appear as one-time buyers, indicating that each customer has only a single recorded transaction.

Insight:

The dataset does not contain repeat purchase information, limiting the ability to analyze customer retention or purchase frequency patterns.

Business Impact:

Due to the absence of repeat transaction data, customer lifetime value and retention strategies cannot be evaluated. Additional data tracking multiple purchases per customer would be required for deeper behavioral analysis.

Conclusion & Key Takeaways

This analysis highlights that revenue is highly concentrated among a small segment of customers and specific product categories. Discounts influence customer spending behavior, while shipping preferences indicate that faster delivery correlates with higher purchase value.

Additionally, older customer segments contribute significantly more to revenue, suggesting targeted marketing opportunities.

However, the dataset lacks repeat purchase data, limiting the ability to analyze customer retention and lifetime value.

Overall, the project demonstrates how data-driven insights can support strategic decision-making in marketing, customer segmentation, and revenue optimization.